

# Comptia A+ Assignment

## [ A+ - Troubleshooting and Helpdesk]

### **BIOS and POST Troubleshooting.**

**Understanding BIOS and POST errors. Analyze POST error codes on a given setup.**

#### Task

- ❖ **1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.**
- When a computer starts, the BIOS performs a POST to check whether the hardware components are working correctly. If a problem is detected, the BIOS displays an error message or beep code to help identify the faulty hardware or incorrect configuration.

| <b>BIOS / POST Error Message</b>       | <b>Hardware or Configuration Issue</b>                                                   |
|----------------------------------------|------------------------------------------------------------------------------------------|
| CMOS Checksum Error                    | The CMOS settings are corrupted or the CMOS battery is weak and may need to be replaced. |
| Keyboard Error or No Keyboard Detected | The keyboard is disconnected, faulty, or the USB/PS/2 port is not functioning properly.  |

|                                                 |                                                                                                                         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| No Boot Device Found / Boot Device Not Detected | The BIOS cannot detect the hard drive or SSD, or the boot order is configured incorrectly.                              |
| Memory (RAM) Error / Memory Test Failed         | The RAM module is loose, damaged, incompatible, or not installed correctly.                                             |
| CPU Fan Error                                   | The CPU cooling fan is not connected, has failed, or is spinning below the required speed, which may cause overheating. |

❖ **2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.**

- In an AMI BIOS, a 3-beep repeating pattern usually indicates a memory (RAM) error. This means the BIOS is unable to detect or properly initialize the system memory during the POST. As a result, the computer cannot continue the boot process, and no display appears on the monitor.
- Troubleshooting Steps:
- Reseat the RAM Module:
  - Turn off the computer and disconnect the power supply.
  - Remove the RAM module, clean the contacts if necessary, and firmly reinstall it into the correct memory slot.

➤ Test or Replace the RAM:

- Try booting the PC with one RAM module at a time (if multiple modules are installed).
- If the problem continues, replace the faulty RAM module with a compatible one.

➤ After correctly reinstalling or replacing the faulty RAM, the POST beep code should stop, and the computer should boot normally with the display appearing on the screen.

❖ **3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.**

➤ To access the BIOS Setup Utility, identify the motherboard manufacturer and BIOS version, and verify the system firmware information.

➤ Step-by-Step Procedure

➤ Step 1: Restart the Computer:

- Restart the PC or Virtual Machine.

➤ Step 2: Enter BIOS Setup:

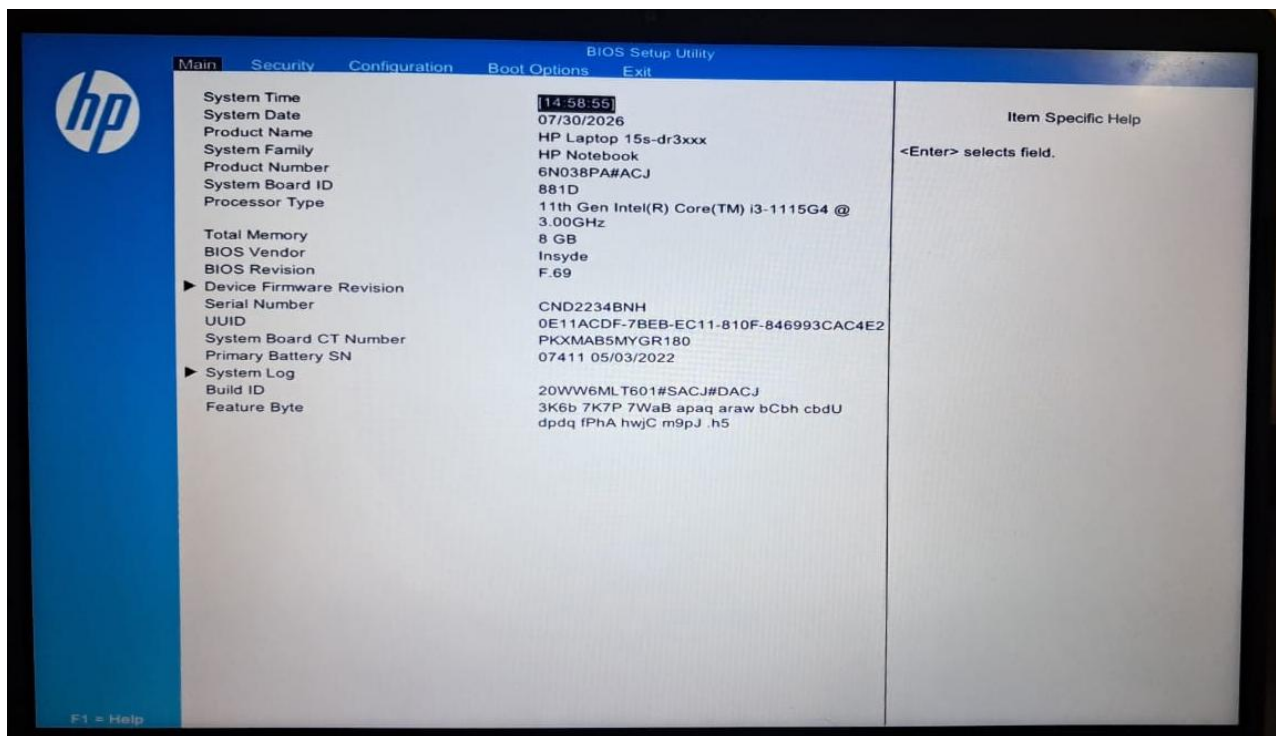
- While the computer is starting, press the BIOS key such as Delete (Del), F2, F10, or Esc depending on the motherboard manufacturer.

➤ Step 3: Open the Main BIOS Screen:

- After entering the BIOS Setup Utility, open the Main or System Information tab.

➤ Step 4: Identify System Information:

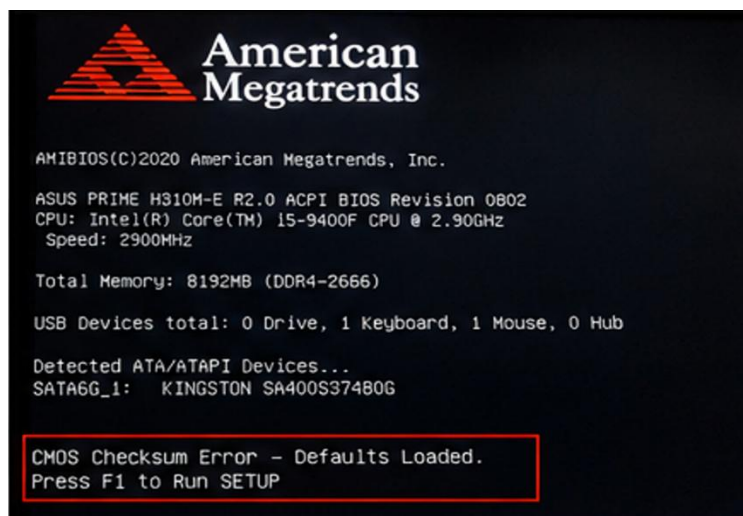
- Motherboard Manufacturer
- BIOS Version
- BIOS Date (if available)



Main BIOS Setup Utility screen showing the motherboard manufacturer and BIOS version.

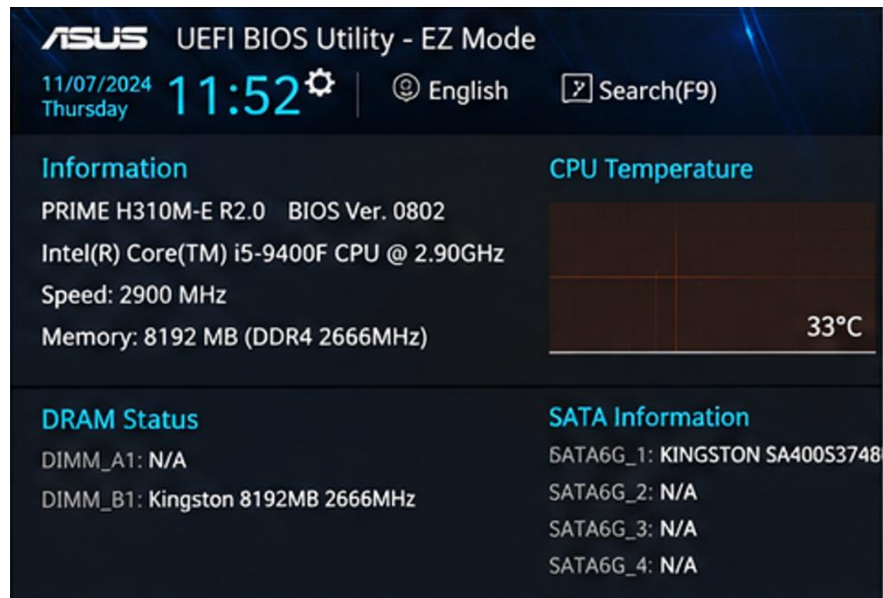
❖ 4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Think about the CMOS battery and BIOS settings reset.

- To understand the CMOS Checksum Error and learn how to troubleshoot and resolve it using BIOS settings and the CMOS battery.
- The "CMOS Checksum Error – Defaults Loaded" message means that the BIOS has detected that the information stored in the CMOS memory is incorrect or has been reset. This usually happens because the CMOS battery (CR2032) is weak or empty, causing the BIOS settings such as date, time, and boot order to return to their default values.
- Step 1: Observe the Error Message:



The BIOS displays the message "CMOS Checksum Error – Defaults Loaded," indicating that the saved BIOS settings have been reset to their default values

➤ Step 2: Enter the BIOS Setup Utility:



Restart the computer and immediately press the Delete (Del), F2, or the appropriate BIOS key to open the BIOS Setup Utility.

➤ Step 3: Check and Correct the BIOS Settings:





Verify the following settings:

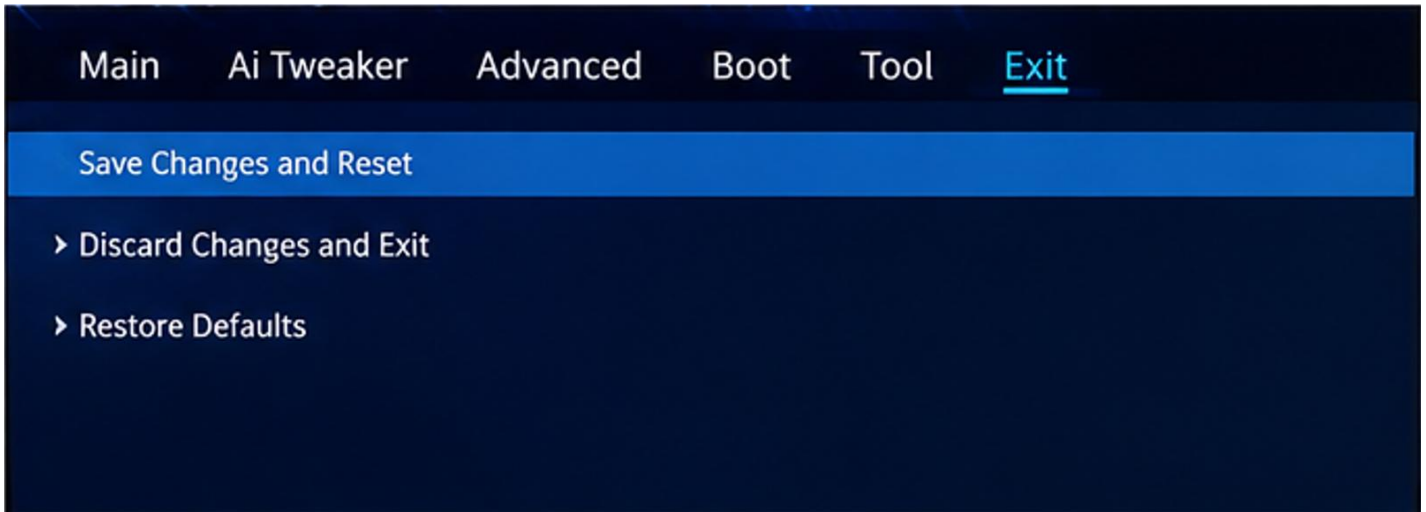
- System Date and Time
- Boot Order
- Storage Devices
- Other BIOS Configuration Settings
- If any settings are incorrect, update them with the correct values.

➤ Step 4: Replace the CMOS Battery (If required):



If the same error appears every time the computer starts, turn off the PC, disconnect the power cable, and replace the CR2032 CMOS battery on the motherboard with a new one.

➤ Step 5: Save Changes and Restart:



Press F10 (Save & Exit) to save the BIOS settings and restart the computer. Verify that the error message no longer appears.

➤ Expected Result:

- After replacing the CMOS battery (if necessary) and saving the correct BIOS settings, the "CMOS Checksum Error – Defaults Loaded" message should disappear, and the computer should boot normally while keeping the correct BIOS configuration.





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